

POLARIS ENERGY TECHNOLOGY RESEARCH INSTITUTE

Work Plan for Inviting External Experts to Conduct Project Technical Consultation

Research Project Management Department

Doc.No.:PRI-2012-EXT-039 | Prepared:September 20,2012 | Dept.:Research Project Management |
Classification:Internal

I. Background and Purpose

According to the phased progress requirements of the "Advanced Reservoir Characterization and Enhanced Oil Recovery Technology Research" project (Project No.:PRI-2012-RES-015) undertaken by our institute, the project team faces technical bottlenecks in key areas such as optimization of reservoir numerical simulation methods, formulation of unconventional oil and gas resource development strategies, and calibration of chemical flooding technology parameters. To ensure the quality and progress of the project research, the Institute Affairs Committee has approved the invitation of internationally renowned experts to conduct a one-week technical consultation and academic exchange activity at our

The objectives of this technical consultation are: (1) to obtain cutting-edge professional guidance on the core technical challenges currently faced by the project; (2) to learn from international advanced research methods and engineering experience to optimize the project technical plan; and (3) to expand international cooperation channels and lay the foundation for subsequent joint

II. Invited Expert Information

Expert Name: Dr. Brenda Allbright **Gender:** Female **Nationality:** Canada
Affiliation: Alberta Energy Research Alliance **Title/Position:** Lecturer **Specialty:** Reservoir Engineering, Enhanced Oil Recovery Technology, Unconventional Oil and Gas Development

Academic Background: Dr. Allbright has published more than 60 research papers in internationally authoritative academic journals and has led multiple major scientific research

projects funded by the Natural Sciences and Engineering Research Council of Canada (NSERC). She possesses profound academic expertise and rich engineering practical experience in the fields of reservoir numerical simulation, chemical flooding enhanced oil recovery technology, and tight reservoir development strategies, and is one of the internationally renowned academic leaders in this

III. Visit Time and Location

Visit Time: October 14, 2012 (Sunday) to October 19, 2012 (Friday), totaling six days and five

Visit Location: Houston, Texas, USA

IV. Detailed Schedule

(1) October 14 (Sunday)

The expert arrives in Houston. The Administrative Affairs Department will arrange airport pickup service and escort her to the accommodation to complete check-in procedures. No formal work arrangements are scheduled for the day, allowing the expert to rest and adjust to the time

(2) October 15 (Monday)

Morning 9:00-12:00: Project Overview Presentation Meeting. Dr. Wang Zhihua, the project leader, will provide the expert with a comprehensive introduction to the project research background, overall technical route, current research progress, and main technical difficulties. Attendees include all core members of the project

Afternoon 14:00-17:00: Laboratory Visit and Technical Document Review. The expert will be arranged to visit our institute's reservoir physical simulation laboratory and numerical simulation computing center, and all project technical documents will be provided for the expert's

(3) October 16 (Tuesday)

Full Day 9:00-17:00: Thematic Technical Seminar (Session 1)—"Optimization of Reservoir Numerical Simulation Methods". The project team will conduct in-depth discussions with the expert on technical issues such as grid division accuracy, history matching methods, and uncertainty quantification in current numerical simulation. Team members will report on technical difficulties in their respective modules, and the expert will provide comments and improvement suggestions one by

(4) October 17 (Wednesday)

Morning 9:00-12:00:Thematic Technical Seminar (Session 2)—"Development Strategies for Unconventional Oil and Gas Resources".Focused discussions on multi-stage fracturing optimization design for horizontal wells in tight reservoirs,shale gas reservoir productivity evaluation methods,and other

Afternoon 14:00-17:00:Internal Project Team Technical Plan Discussion Meeting.Based on the results of the previous two days's eminars,the project team will revise and discuss the technical plan,with the expert participating throughout and providing

(5)October 18(Thursday)

Full Day:On-site Technical Inspection.The expert will be organized to visit our institute's cooperative oilfield site (Eagle Ford Block Operation Area)for field research,inspecting on-site production data collection systems,injection-production process facilities,and conducting technical exchanges with on-site engineering and technical personnel.Transportation for the day will be arranged uniformly by the

(6)October 19(Friday)

Morning 9:00-12:00:Comprehensive Discussion and Technical Summary Meeting.The expert will provide a systematic summary of the technical issues covered during this consultation period,forming preliminary consultation

Afternoon 14:00-16:00:Follow-up Cooperation Intention Discussion.Both parties will conduct preliminary consultations on possible future joint research directions,personnel exchange plans,and results sharing

V.Reception Arrangements

(1)Accommodation Arrangements

Executive-level guest room accommodation will be arranged for the expert at our institute's contracted cooperative hotel,with the stay period from October 14to October 19,totaling five room nights.Reasonable incidental expenses incurred during the stay (including in-hotel dining,parking,etc.)will be included in this reception expense and consolidated for

(2)Transportation Arrangements

Responsible for the expert's airport pickup service on October 14and airport drop-off service on October 19,as well as full transportation support during the on-site inspection on October 18.Valet parking and parking fees incurred during the expert's visit will be included in the reception expense for unified

(3)Catering Arrangements

During the workdays from October 15 to October 19, lunch will be uniformly arranged at the institute's staff cafeteria (standard work meals). Other meals will be arranged by the expert independently, with related expenses executed according to our institute's travel reimbursement management

VI. Work Output Requirements

1. 1. The expert must submit one written technical consultation report by October 20, covering evaluation opinions on each technical aspect of the project, improvement suggestions, and recommended follow-up research directions;
2. 2. The project team must complete the minutes organization and archiving work within two weeks after the expert's departure (i.e., before November 2, 2012);
3. 3. The Research Project Management Department is responsible for incorporating the key recommendations from the technical consultation report into the revised project implementation plan, which will be submitted to the Institute Affairs Committee for approval and execution;
4. 4. All visit archive materials (including work plans, approval documents, meeting records, consultation reports, expense reimbursement vouchers, etc.) will be uniformly archived by the Research Project Management Department, with a retention period of no less than five

VII. Budget Management

All expenses related to this expert visit will be strictly executed in accordance with our institute's "Travel and Business Entertainment Expense Reimbursement Management System" (POL-FIN-2011-003). Expense categories include accommodation expenses (including reasonable incidental expenses), transportation expenses (including valet parking fees), catering expenses, and other reasonable expenditures approved through the approval process. All expenses will be based on actual amounts incurred, reimbursed with compliant invoices, and reimbursement materials must be accompanied by this work plan and approval documents as proof of

VIII. Responsibility Division

5. 1. Research Project Management Department (Zhang Weiqiang): Responsible for overall coordination, schedule arrangement, technical data preparation, and archive management;
6. 2. Reservoir Technology Research Laboratory (Wang Zhihua): Responsible for preparation of technical discussion content, on-site inspection arrangement, and consultation report receipt;

7. 3.Administrative Affairs Department (Zhou Liping):Responsible for accommodation booking,transportation support,catering arrangement,and logistics services;
8. 4.Finance Department (Zhao Haiyan):Responsible for expense review and reimbursement

IX.Contingency Plan

9. 1.If the expert is unable to arrive on time due to visa,flight,or other reasons,the Research Project Management Department should promptly communicate with the expert to adjust the schedule and notify relevant departments;
10. 2.If force majeure factors prevent the on-site inspection,the October 18schedule may be adjusted to supplementary in-house technical seminars;
11. 3.All adjustment items must be approved by the department director before execution,and written records must be

Prepared by:

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